Concept Doc: Habit Replacement Engine

Working title: TBD (candidates: Rewire, Outgrow, Seedless, Overwrite) **Company:** Artomai **Status:** Pre-MVP concept · v0.1

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1. One-Line Thesis

A private, stigma-free habit-replacement engine that uses the mechanics of dependency (variable rewards, ritual, identity) to outcompete unwanted habits — starting with cannabis, generalizing to any compulsive behavior.

2. The Problem

- Only ~7% of people with cannabis use disorder receive treatment. The top reasons for not seeking help: stigma, cost, preference to quit alone, and belief that formal treatment is "too much."
- Existing quit apps are streak counters dressed in recovery language. They ask users to self-identify as addicts on day one — the exact framing that keeps the target user away.
- Willpower-based abstinence fails because addiction is a learned reward loop. The loop doesn't get deleted; it gets outcompeted.

Target user: The private struggler. Employed, functional, ashamed, has tried quitting multiple times, will never walk into a group or a therapist's office, but wants out.

3. Positioning

Not a recovery app. A training program you finish.

- No addiction language in the UI. Habits are "programs." Quitting weed sits next to quitting sugar, doomscrolling, nicotine, late-night gaming.
- Privacy is structural, not a feature: nobody can tell which program you're running.
- The graduation frame: the app deliberately weans you off itself. Finishing is the flex.
- Multi-habit design is the retention model: succeed at one program → point the engine at the next. Retention through graduation, not relapse.

4. Core Loop

1. **Map** — Onboarding builds a personal trigger map: times, moods, places, rituals tied to the habit.
2. **Catch** — User logs a craving the moment it hits (one tap, <2 seconds).
3. **Ride** — Guided urge-surf: a 10–20 minute timer with a micro-activity, because cravings peak and decay fast.
4. **Replace** — A habit-specific substitute ritual fills the exact cue slot (not generic "go for a walk").
5. **Reward** — Variable-ratio reward fires on the *beaten craving*, not the daily check-in. Beating a craving is the jackpot moment.
6. **Compound** — Visible accumulation: money saved, sleep score, clarity days, a growing world/avatar. Loss aversion protects the streak.
7. **Taper** — Reward density front-loads in the first 72 hours (highest relapse risk), then tapers as real-world rewards come online. Graduation is the final unlock.

5. Reward Economy (the science-backed part)

Mechanic	Basis	Implementation
Variable-ratio rewards	Operant conditioning; strongest schedule for behavior maintenance	Randomized unlocks/bonuses on craving wins
Contingency management (CM)	One of the most effective SUD interventions across 100+ RCTs; rewarding verified abstinence works	Escalating rewards for consecutive clean days; reset rules that punish lightly, reward recovery fast
Escalating + reset structure	Core CM principle: larger rewards for longer abstinence streaks	Streak multipliers with a "soft reset" (relapse costs the multiplier, not the world)
Front-loaded density	Most relapse occurs in the first days/weeks	72-hour "surge mode" with high reward frequency
Identity progression	Identity change predicts lasting behavior change	Profile evolves as programs complete; user becomes "a person who rewires habits"

Implementation intentions	If-then planning roughly doubles follow-through in behavior change literature	Trigger map generates personalized if-then plans automatically

Design guardrail: the app must be a scaffold, not a new cage. Taper is not optional; it is the product's moral spine and its marketing story.

6. Hardening: Risks & Open Questions

Efficacy risks

- Digital-intervention effect sizes for cannabis are real but *small* in meta-analyses, and at least one strong RCT (ICan, 2023) failed to beat an educational control. Lesson: content modules alone don't work — the differentiators must be craving-moment intervention + CM-style rewards, which most tested apps lacked.
- CM's known weakness: effects decay after rewards stop. The taper design and identity mechanics exist specifically to counter this. This should be our core research question.
- Self-report is gameable. For v1 that's acceptable (the user is only cheating themselves), but consider optional verification later (CM apps like DynamiCare use biomarker testing — likely overkill for our positioning).

Product risks

- Craving logging requires the user to open the app *during* the hardest moment. Friction must be near zero: widget, lock-screen button, watch complication.
- Withdrawal is real (sleep disruption, irritability, vivid dreams for 1–4 weeks). The app needs content for this window or users will interpret withdrawal as "the app isn't working."
- Regulatory line: stay a *wellness product*, not a medical device. No diagnosis, no treatment claims, no "cures addiction." Language review before launch.
- Duty of care: detect crisis signals (self-harm language in journals) and surface professional resources gracefully without breaking the no-stigma promise.

Missed opportunities to explore

- **Money-saved as the hero metric.** Cannabis is expensive; a live "reclaimed money" counter that can be pledged toward a real purchase is CM with the user's own money.
- **Sleep/clarity integration.** Pull HealthKit/Google Fit sleep data — showing *measured* improvement is the strongest retention hook and no competitor does it well.

- **Anonymous cohort effects.** Not community (breaks privacy promise), but "3,412 people beat a craving today" — social proof without identity.
- **B2B2C later:** employers/insurers pay for CM-style programs; the destigmatized framing makes corporate wellness distribution plausible where "addiction apps" die.
- **Content packs as the expansion model:** engine is universal, packs are habit-specific (weed, nicotine, sugar, doomscroll, alcohol-moderation*). *Alcohol needs extra safety design — cold-turkey withdrawal can be medically dangerous; the app must screen and redirect heavy alcohol cases.

7. Building on Real Science: How to Find Unbiased Studies

Where to look (in order of trust):

1. **Cochrane Library** (cochranelibrary.com) — gold-standard systematic reviews; strict bias-assessment methodology; independent of industry.
2. **PubMed / PubMed Central** (pubmed.ncbi.nlm.nih.gov) — filter by "systematic review," "meta-analysis," and "randomized controlled trial." Free full text on PMC.
3. **ClinicalTrials.gov** — see *pre-registered* trials, including ones that failed (protects against publication bias). Search "cannabis use disorder" + "digital."
4. **PROSPERO** (crd.york.ac.uk/prospero) — registry of systematic review protocols; find reviews before they're spun.
5. **Google Scholar** — for citation-chasing: find one good meta-analysis, then follow who cites it.

How to judge bias (our internal checklist):

- Pre-registered? (Trial registered before data collection = harder to cherry-pick outcomes)
- Who funded it? (Check conflict-of-interest statements; be wary of app-vendor-funded efficacy studies — including, eventually, our own)
- RCT with an *active* control? (Beating "no treatment" is easy; beating an educational control is the real bar — see ICan)
- Sample size and follow-up length? ($n<100$ or <3-month follow-up = suggestive, not load-bearing)
- Effect size reported, not just p-values? ($d \approx 0.2$ small, 0.5 medium, 0.8 large)
- Published in a journal with real peer review? (Addiction, JAMA Psychiatry, Drug and

Alcohol Dependence, Journal of Substance Abuse Treatment, JMIR)

Priority reading list to seed the algorithm design:

- Boumparis et al. 2019 — meta-analysis of 21 RCTs on digital cannabis interventions (what works, what's marginal)
- Olthof et al. 2023 (ICan RCT, *Addiction*) — the cautionary tale: well-built guided app that didn't beat educational control
- Prendergast et al. 2006 + the 2021 JAMA Psychiatry CM meta-analysis — why contingency-style rewards are the strongest single mechanism
- Literature on "Quit the Shit" (Germany) and "Reduce Your Use" (Australia) — the two most-studied cannabis programs; steal what worked
- Behavior change technique (BCT) taxonomy work (Michie et al.) — the component-level map of which techniques drive effects, ideal for structuring our engine

Long-term credibility play: run our own pre-registered pilot study (partner with a university lab; they get data and publications, we get independent validation). This is the moat no competitor quit-app has.

8. MVP Scope (v1, one habit: cannabis)

In:

- Trigger-mapping onboarding (10 min, no addiction language)
- One-tap craving log + urge-surf timer + replacement ritual prompt
- Variable reward engine with escalating streaks and soft resets
- Money-saved + clarity-days compounding dashboard
- 30-day "program" arc with 72-hour surge mode and withdrawal-window content
- Fully anonymous account option

Out (v2+):

- Additional habit packs, health-data integration, cohort stats, graduation certification, B2B channel

Success metrics for the pilot:

- D7 / D30 retention vs. quit-app benchmarks
- % of logged cravings marked "beaten"

- Self-reported use-days at day 30 vs. baseline (the research-standard outcome)

Next steps: name + brand direction, wireframes of the core loop, and a lit-review sprint through the priority reading list to lock the reward-schedule parameters.